

Cybersecurity Training Diary

Day 13: Scanning & Enumeration

1. Topics Covered

✓ Network Scanning Techniques

Type	Purpose	Tool Example	Command
Ping Sweep	Find live hosts	<code>fping</code>	<code>fping -g 192.168.1.0/24</code>
Port Scan	Discover open ports	<code>nmap</code>	<code>nmap -sS -T4 192.168.1.1</code>
OS Detection	Identify operating systems	<code>nmap -O</code>	<code>nmap -O 192.168.1.1</code>
Service Scan	Detect running services	<code>nmap -sV</code>	<code>nmap -sV -p 80,443 192.168.1.1</code>

✓ Enumeration Methods

```
graph LR
    A[Enumeration] --> B[SMTP]
    A --> C[SMB]
    A --> D[SNMP]
    A --> E[HTTP]
    B --> B1[VRFY Command]
    C --> C1[Null Sessions]
    D --> D1[Public Strings]
    E --> E1[Directory Busting]
```

✓ Vulnerability Correlation

- **Open Port 445** → Potential SMB exploits
- **HTTP Server Banner** → Apache 2.4.7 (CVE-2017-15715)
- **FTP Anonymous Login** → File extraction risk

2. Key Takeaways

- ✓ **Stealth vs Speed:** SYN scans (`-sS`) are stealthier than TCP connects (`-sT`)
- ✓ **Enumeration is key:** Discover usernames (email formats), shares, apps
- ✓ **Time management:** Slow scans (`-T2`) evade detection but take longer

3. Practical Exercises

♦ Exercise 1: Full Nmap Scan

```
nmap -sS -sV -O -p- -T4 -oA full_scan 192.168.1.1
# -sS: SYN scan
# -p-: All ports
# -oA: Output all formats
```

♦ Exercise 2: SMB Enumeration

```
enum4linux -a 192.168.1.100
# Extract user lists, shares
```

4. Questions for Reflection

- ? Why would an attacker scan UDP ports (like 161/SNMP)?
- ? How could you detect scanning activity on your network?